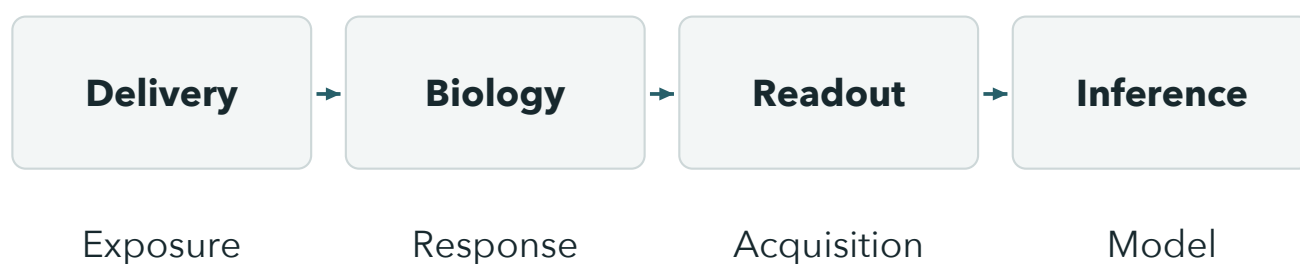


Therapeutics, investigations and quantitative evidence

Part 1 | Evidence links exposure to a measured response

A treatment effect, a test result and an analysis each have a different chain of assumptions.



Distinct failure points require distinct controls

Delivery, biological response, measurement and inference are connected; a failure can occur at any stage.

Question

Feature	Necessary distinction
Did drug reach tissue?	Dose versus exposure
Did biology change?	Exposure versus response
Did the test capture it?	Response versus measurement
Is the estimate valid?	Measurement versus design assumptions

Pharmacokinetics separates amount from replacement rate

Distribution and clearance govern different dosing quantities.

Original linear model; $F=1$ and target 1 mg/L

Variable	Baseline	After change
Vd	10 L	20 L
Clearance	2 L/h	1 L/h
Loading dose	10 mg	20 mg
Maintenance rate	2 mg/h	1 mg/h

Why the changes differ

More apparent distribution volume requires more initial drug to reach the target concentration. Lower clearance requires less continuing replacement to maintain it. These simplified results assume stable linear kinetics and comparable bioavailability; they are explanatory calculations rather than a clinical dosing plan.

Investigation choice follows the generator and specimen

Different assays can agree or disagree because they measure different levels.

Comparison

Feature	What differs	Useful conclusion
PERG versus VEP	Retinal versus cortical recording	Localisation needs controlled input
Culture versus PCR	Viable growth versus target DNA	Discordance can reflect treatment
DNA versus RNA assay	Sequence versus transcript	Unchanged sequence does not prove unchanged expression

Acquisition validity

Defocus can alter a pattern-dependent response; reaction inhibition can invalidate a molecular negative; poor sampling can undermine both microbiological methods. A validated measurement requires controls appropriate to the mechanism of acquisition, rather than a generic assumption that every reported negative is equivalent.

Diagnostic probability and trial uncertainty are different

A positive predictive value updates disease probability; a confidence interval describes estimate uncertainty under a model.

Evidence question

Feature	Relevant quantity	Dependence
Probability after positive test	Predictive value	Pretest probability and test performance
Threshold trade-off	Sensitivity/specificity	Chosen operating point
Uncertainty in two-eye trial	Correct standard error	Fellow-eye correlation
Noninferiority	Interval versus margin	Prespecified clinical boundary

Integrated reading

Lower prevalence can reduce PPV while fixed sensitivity and specificity remain unchanged. Ignoring positive eye correlation can narrow an interval without reversing its point estimate. Different errors affect different parts of the inference; identify the assumption that actually changed.

Mechanistic reasoning preserves what was controlled

A useful conclusion explains the changed outcome while respecting the preserved processes.

Controlled contrasts

Experiment	Preserved versus changed	Inference
Retinal response	Normal onset; delayed recovery	Shut-off/restoration
Lens optics	Same curvature; more scatter	Microscopic organisation
PCR reaction	Control fails; target negative	Negative uninterpretable
Pedigree	Two carriers; offspring unaffected	Conditional denominator

A completed synthesis

Controls narrow alternatives; they do not magically prove a unique mechanism. The strongest answer fits the biological level, direction of change and experimental boundary. A clinical label is less informative than a causal explanation when the question deliberately supplies a perturbation and a measured response.